

Chapter 6

Continuous Math I: Interest and Loans

Money growing over time is governed by the same two recursions introduced in Chapter 1. Simple interest applies the additive recursion: the same fixed amount is added each period. Compound interest applies the multiplicative recursion: the balance is multiplied by the same factor each period, so interest earns interest. A loan places these two recursions in tension: compound interest accrues on the outstanding balance while fixed payments reduce it. This chapter develops each model, connects them to the recursion framework, and examines the financial consequences of each. Throughout, treat the balance as a changing state: ask what is added or multiplied, what acts against the change, and how that update shapes the next period.

A continuous view. The calculations below update balances by year or month. A smooth exponential model expresses the same proportional-growth pattern continuously: the rate of change is proportional to the current balance, written $\frac{dB}{dt} = kB$ with solution $B(t) = B_0e^{kt}$.

6.1 Simple Interest

Definition. *Simple interest.* **Simple interest** is calculated on the original principal only. The interest earned over t years at annual rate r is:

$$I = P \cdot r \cdot t,$$

and the total amount after t years is:

$$A = P + I = P(1 + rt),$$

where P is the principal, r is the annual interest rate expressed as a decimal, and t is time in years.

Simple interest is the additive recursion from Chapter 1. Each year, the same fixed amount $P \cdot r$ is added to the balance. In state-update notation, $B_{n+1} = B_n + Pr$. The current

balance does not change that year's interest amount. This is additive change, represented by a straight line when graphed.

Example 6.1.1. You deposit \$2,000 in an account that pays 3% simple interest per year.

- (a) How much interest is earned each year?

$$I_{\text{annual}} = 2000 \times 0.03 = \$60.$$

- (b) What is the balance after 5 years?

$$A = 2000(1 + 0.03 \times 5) = 2000(1.15) = \$2,300.$$

- (c) Fill in the balance for each year.

Year	Balance
0	\$2,000
1	\$2,060
2	\$2,120
3	\$2,180
4	\$2,240
5	\$2,300

Each row adds \$60: this is the additive recursion with step size \$60, starting from \$2,000.

Example 6.1.2. (*Finding the rate*) An account grows from \$500 to \$560 in 4 years under simple interest. What is the annual interest rate?

$$560 = 500(1 + r \times 4).$$

$$\frac{560}{500} = 1 + 4r \Rightarrow 1.12 = 1 + 4r \Rightarrow 4r = 0.12 \Rightarrow r = 0.03 = 3\%.$$

Example 6.1.3. (*Finding the time*) At what year does a \$1,000 account earning 5% simple interest reach \$1,250?

$$1250 = 1000(1 + 0.05t) \Rightarrow 1.25 = 1 + 0.05t \Rightarrow 0.05t = 0.25 \Rightarrow t = 5 \text{ years}.$$

Exercises 6.1

6.1.1. Compute the total amount A for each situation.

- (a) $P = \$1,500$, $r = 4\%$, $t = 3$ years
- (b) $P = \$800$, $r = 6\%$, $t = 5$ years
- (c) $P = \$3,000$, $r = 2.5\%$, $t = 8$ years

6.1.2. An account earns 5% simple interest per year. The starting balance is \$1,200.

- (a) Fill in the balance table for years 0 through 6.
- (b) By how much does the balance increase each year?
- (c) Write the balance at year t as a formula.

6.1.3. An account grows from \$4,000 to \$4,600 in 5 years under simple interest. Find the annual interest rate.

6.1.4. At 4% simple interest, how many years does it take a \$2,500 account to reach \$3,000?

6.2 Compound Interest

Definition. *Compound interest.* **Compound interest** is calculated on the current balance, including previously earned interest. The amount after t years at annual rate r , compounded once per year, is:

$$A = P(1 + r)^t,$$

where P is the principal and r is the annual rate as a decimal.

Compound interest is the multiplicative recursion from Chapter 1. Each year the current balance is multiplied by $(1 + r)$: $B_{n+1} = (1 + r)B_n$. The current state feeds the next interest calculation, so each increase can make later increases larger. This is an amplifying feedback pattern and produces an upward-bending exponential curve.

Example 6.2.1. You deposit \$2,000 at 3% interest, compounded annually.

- (a) What is the balance after 5 years?

$$A = 2000(1.03)^5 = 2000 \times 1.1593 \approx \$2,318.55.$$

- (b) Fill in the balance for each year by multiplying the previous balance by 1.03.

Year	Balance	Interest earned
0	\$2,000.00	
1	\$2,060.00	\$60.00
2	\$2,121.80	\$61.80
3	\$2,185.45	\$63.65
4	\$2,251.02	\$65.57
5	\$2,318.55	\$67.53

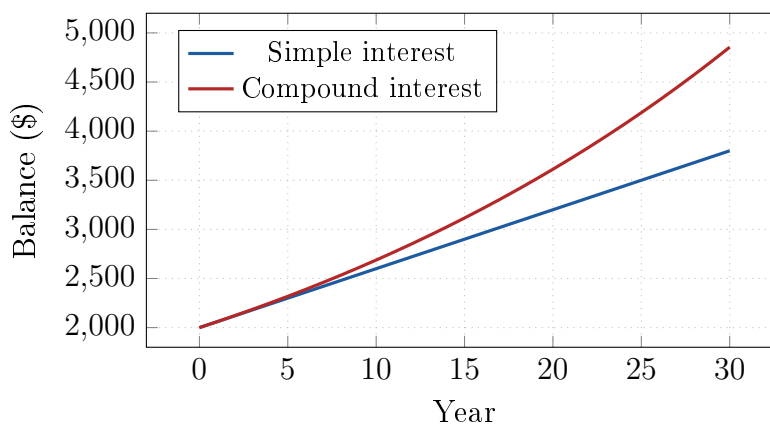
The interest earned increases each year because the base on which it is calculated grows. This is the key distinction from simple interest, where the interest earned is the same every year.

Example 6.2.2. (*Long-term growth*) Compare the two accounts from Example 6.1.1 and Example 6.2.1, both starting at \$2,000 at 3% over 30 years.

$$A_{\text{simple}} = 2000(1 + 0.03 \times 30) = 2000(1.90) = \$3,800.$$

$$A_{\text{compound}} = 2000(1.03)^{30} \approx 2000 \times 2.4273 = \$4,854.52.$$

The compound account produces \$1,054.52 more over 30 years.



Compounding More Frequently

Interest can be compounded more than once per year. If interest is compounded n times per year, each compounding period uses the rate $\frac{r}{n}$, and there are nt compounding periods in t years.

Definition. *General compound interest formula.*

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

where n is the number of compounding periods per year.

Example 6.2.3. \$1,000 at 6% annual interest for 10 years under different compounding frequencies.

Compounding	n	Balance after 10 years
Annually	1	\$1,790.85
Semi-annually	2	\$1,806.11
Quarterly	4	\$1,814.40
Monthly	12	\$1,819.40
Daily	365	\$1,822.03

More frequent compounding produces a higher balance, but the differences diminish as the frequency increases. The balance is updated more often, and each percentage is applied to the latest state.

Exercises 6.2

6.2.1. Compute the total amount A for each situation. Round to the nearest cent.

- (a) $P = \$1,000$, $r = 5\%$, $t = 4$ years, compounded annually
- (b) $P = \$500$, $r = 8\%$, $t = 3$ years, compounded annually
- (c) $P = \$2,000$, $r = 4\%$, $t = 6$ years, compounded monthly

6.2.2. An account starts with \$3,000 at 4% compounded annually.

- (a) Fill in the balance table for years 0 through 5 by multiplying each year's balance by 1.04.
- (b) Compare the interest earned in year 1 with the interest earned in year 5. Explain the difference in one sentence.

6.2.3. \$1,500 is invested at 6% for 10 years. Compute the final balance under:

- (a) Simple interest

(b) Compound interest, compounded annually

(c) Compound interest, compounded monthly

Rank the three results from smallest to largest.

6.2.4. A savings account compounds monthly at an annual rate of 3%. What is the effective annual yield? Use the formula $A = P(1 + r/12)^{12}$ with $P = 1$ and $t = 1$, and subtract 1 from the result.

6.3 The Rule of 72

The Rule of 72 is an approximation for the doubling time of a compound interest account.

Definition. *Rule of 72.* At an annual compound interest rate of $r\%$, money approximately doubles in:

$$t \approx \frac{72}{r} \text{ years.}$$

The Rule of 72 applies equally to any quantity growing by a fixed percentage per period: savings, debt, population, or inflation. It gives a quick view of how a repeated multiplicative update amplifies a starting state over time.

Example 6.3.1.

- (a) At 6% annual interest, money doubles in approximately $72 \div 6 = 12$ years.
- (b) At 9%, money doubles in approximately $72 \div 9 = 8$ years.
- (c) A credit card charges 18% annual interest. An unpaid balance doubles in approximately $72 \div 18 = 4$ years if no payments are made.
- (d) Inflation at 3% per year halves the purchasing power of money in approximately $72 \div 3 = 24$ years.

Example 6.3.2. (*Verifying the Rule*) Using the compound interest formula to find the exact doubling time at 6%:

$$2P = P(1.06)^t \Rightarrow 2 = (1.06)^t.$$

The exact solution requires logarithms and gives $t \approx 11.9$ years. The Rule of 72 gives $72/6 = 12$ years, a close approximation.

Exercises 6.3

6.3.1. Use the Rule of 72 to estimate the doubling time for each interest rate.

- (a) 4%
- (b) 8%
- (c) 12%
- (d) 2%

6.3.2. A student loan has an interest rate of 6%. The student does not make payments for 12 years while in graduate school.

- (a) Use the Rule of 72 to estimate whether the loan balance has doubled after 12 years.
- (b) If the original loan was \$30,000, estimate the balance after 12 years.

6.3.3. If money doubles in 9 years under compound interest, what is the approximate annual interest rate? Set up the Rule of 72 equation and solve for r .

6.3.4. An investment grows at 6% per year. Using the Rule of 72, estimate how many years it takes to grow to four times its original value. *Hint: four times means two doublings.*

6.4 Loans

A loan begins with a lender providing a principal P to a borrower. Each period, two things happen.

1. The outstanding balance increases by compound interest.
2. The borrower makes a fixed payment that reduces the balance.

The monthly interest rate is $r_m = r_{\text{annual}} \div 12$. The balance is updated by two competing effects: interest uses the current balance and increases it; the scheduled payment is then subtracted. If the payment is no larger than the interest charged, the balance will not decrease.

The balance after each payment is computed as:

$$B_{n+1} = B_n(1 + r_m) - M, \quad \text{new balance} = (\text{current balance})(1 + r_m) - \text{payment}.$$

Example 6.4.1. A \$4,000 loan at 6% annual interest compounded monthly, with fixed monthly payments of \$200. Monthly interest rate: $r_m = 0.06 \div 12 = 0.005$.

Month	Balance	Interest	After payment
0	\$4,000.00		
1	\$4,000.00	+\$20.00	\$3,820.00
2	\$3,820.00	+\$19.10	\$3,639.10
3	\$3,639.10	+\$18.20	\$3,457.30
4	\$3,457.30	+\$17.29	\$3,274.59
5	\$3,274.59	+\$16.37	\$3,090.96

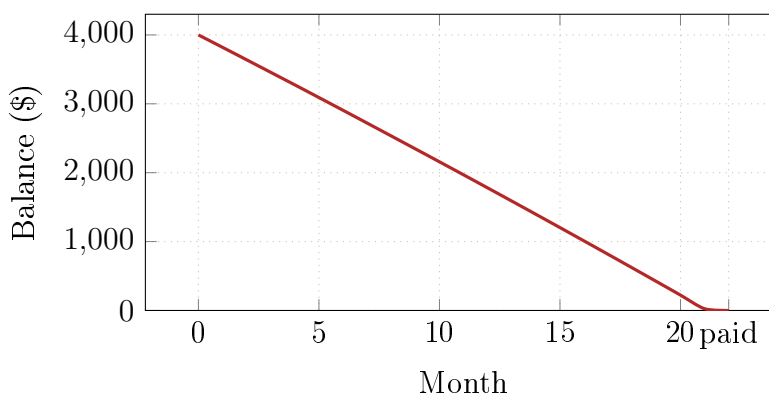
In month 1, \$20 of the \$200 payment goes to interest and \$180 reduces the principal. In month 5, only \$16.37 goes to interest, and \$183.63 reduces the principal. As the balance falls, each payment retires a larger fraction of principal.

Example 6.4.2. (*Total cost of a loan*) Continuing the loan from Example 6.4.1, 21 payments of \$200 leave a balance of about \$24.87 before month 22. After one more month of interest, the final payment is about \$24.99. Therefore,

$$\text{Total paid} = 21(\$200) + \$24.99 = \$4,224.99,$$

$$\text{Total interest} = \$4,224.99 - \$4,000 = \$224.99.$$

The final payment is smaller than the regular payment because it only needs to clear the remaining balance.



Example 6.4.3. (*Effect of payment size*) The same \$4,000 loan at 6% annual interest compounded monthly, under different payment amounts. The final payment may be smaller than the regular payment; totals below use cent-rounded monthly interest.

Monthly payment	Months to pay off	Total paid	Total interest
\$150	29	\$4,303.84	\$303.84
\$200	22	\$4,224.99	\$224.99
\$300	14	\$4,150.01	\$150.01

Larger payments shorten the loan term and reduce total interest paid.

Exercises 6.4

6.4.1. A \$3,000 loan is charged 9% annual interest compounded monthly. The monthly interest rate is $0.09 \div 12 = 0.0075$. The monthly payment is \$150.

(a) Complete the loan table for the first four months.

Month	Balance	Interest	After payment
0	\$3,000.00		
1	\$3,000.00	$\$3,000 \times 0.0075 =$	
2			
3			
4			

(b) In month 1, how much of the \$150 payment goes toward interest, and how much toward reducing the principal?

6.4.2. A car loan of \$12,000 at 4.8% annual interest compounded monthly has a monthly payment of \$400. Monthly rate: $0.048 \div 12 = 0.004$.

(a) Complete the loan table for the first three months.

(b) After the third payment, what fraction of the original principal has been repaid?

6.4.3. A student owes \$8,000 on a credit card charging 18% annual interest compounded monthly (monthly rate = 0.015). The minimum payment is \$160 per month.

(a) Compute the interest charge in month 1.

- (b) How much of the first payment reduces the principal?
- (c) At this rate, explain in one sentence whether the minimum payment is sufficient to eventually pay off the debt.

6.4.4. Explain in two complete sentences why the fraction of each payment going toward interest decreases over the life of a loan. Use the loan balance as part of your explanation.

6.5 The Observer and the Loan

A loan agreement contains a fixed set of numbers: principal, interest rate, payment amount, and term. The bank and the borrower extract different information from those same numbers because they are asking different questions.

The bank asks	The borrower asks
What is the total interest collected?	What is my monthly payment?
What is the annual percentage yield?	How long until I am debt-free?
What is the default risk?	How much of each payment is interest?
What is the effective return on capital?	What does this loan cost in total?

The loan terms and balance remain the same; the bank and borrower choose different quantities to calculate because they are asking different questions. The shared information can support several perspectives, and each perspective selects a different feature to interpret.

This structure also appears in the comparison between simple and compound interest. An observer asking “what is the balance after one year?” gets the same answer from both models at a 3% rate (both earn \$60 on a \$2,000 principal in year one). An observer asking “what is the balance after 30 years?” gets very different answers. The question determines when the distinction matters.

Example 6.5.1. A borrower takes a \$10,000 loan at 5% annual interest compounded monthly. The regular monthly payment is \$230.30 for 47 months, followed by a final payment of about \$229.96 in month 48.

The bank’s perspective:

- Total collected: $47 \times \$230.30 + \$229.96 = \$11,054.06$.
- Total interest: $\$11,054.06 - \$10,000 = \$1,054.06$.
- Total interest is about 10.54% of the original principal over 4 years.

The borrower's perspective:

- The scheduled payment is about \$230.30 per month.
- The loan is retired in 48 months.
- Total cost above principal is \$1,054.06.

Exercises 6.5

6.5.1. A \$6,000 loan at 7.2% annual interest compounded monthly has a regular payment of \$185.82 for 35 months and a smaller final payment in month 36.

- (a) Compute the total amount paid.
- (b) Compute the total interest paid.
- (c) Write one sentence describing this loan from the borrower's perspective.
- (d) Write one sentence describing this loan from the lender's perspective.

6.5.2. Two loans both have a principal of \$5,000 and an annual rate of 6% compounded monthly. Loan A has monthly payments of \$150. Loan B has monthly payments of \$250.

- (a) Without completing full tables, predict which loan will cost more in total interest. Explain in one sentence.
- (b) Compute the first month's interest charge for each loan.
- (c) Explain why both loans have the same first month interest charge despite different payments.

Chapter 6 Review

1. A \$5,000 investment earns 4% simple interest per year.
 - (a) Write the balance at year t as a formula.
 - (b) Fill in the balance for years 0 through 5.
 - (c) How many years until the balance reaches \$6,000?
2. The same \$5,000 investment earns 4% compound interest per year.
 - (a) Fill in the balance for years 0 through 5 by multiplying each year by 1.04.
 - (b) Compare the interest earned in year 1 and year 5.

- (c) How much more does the compound account hold after 5 years than the simple interest account?
3. Use the Rule of 72 for each situation.
- (a) At what annual rate does money double in approximately 6 years?
 - (b) A retirement account earns 7% per year. Approximately when will it double? When will it quadruple?
4. A \$2,500 loan at 12% annual interest compounded monthly (monthly rate = 0.01) has monthly payments of \$120.
- (a) Complete the loan table for months 1 through 4.
 - (b) How much of the month-1 payment reduces the principal?
 - (c) Estimate how many months the loan will take to pay off if the balance decreases by roughly \$95 per month on average.
5. A credit card charges 24% annual interest compounded monthly (monthly rate = 0.02). A balance of \$3,000 accrues with no payments made.
- (a) Use the Rule of 72 to estimate how many years until the balance doubles.
 - (b) Compute the balance after 12 months using the compound interest formula. Does this match the Rule of 72 estimate?
6. Two accounts each start with \$1,000. Account A earns 5% simple interest. Account B earns 4% compounded annually.
- (a) Fill in balances for both accounts at years 5, 10, 20, and 30.
 - (b) At what approximate year does Account B overtake Account A?
 - (c) Explain in two sentences why the compound account eventually exceeds the simple account, even though it has a lower interest rate.
7. A homebuyer takes a mortgage of \$200,000 at 4.5% annual interest compounded monthly. It is to be repaid over 30 years with monthly payments of \$1,013.37.
- (a) Compute the total amount paid over 30 years.
 - (b) Compute the total interest paid.
 - (c) Express the total interest as a percentage of the original loan amount.
 - (d) Write two sentences: one from the borrower's perspective and one from the bank's perspective on this mortgage.